Presentation

On

Title name

Deep Forensic Model with Generative AI for Image Forgery Detection and Localization

Group Number: 17

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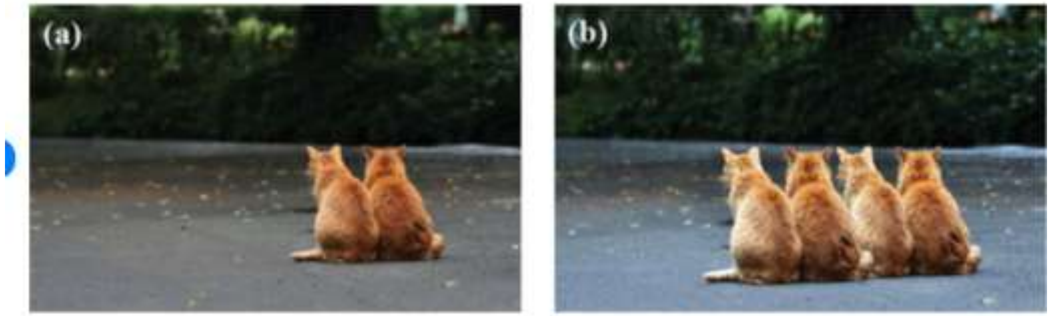
Introduction

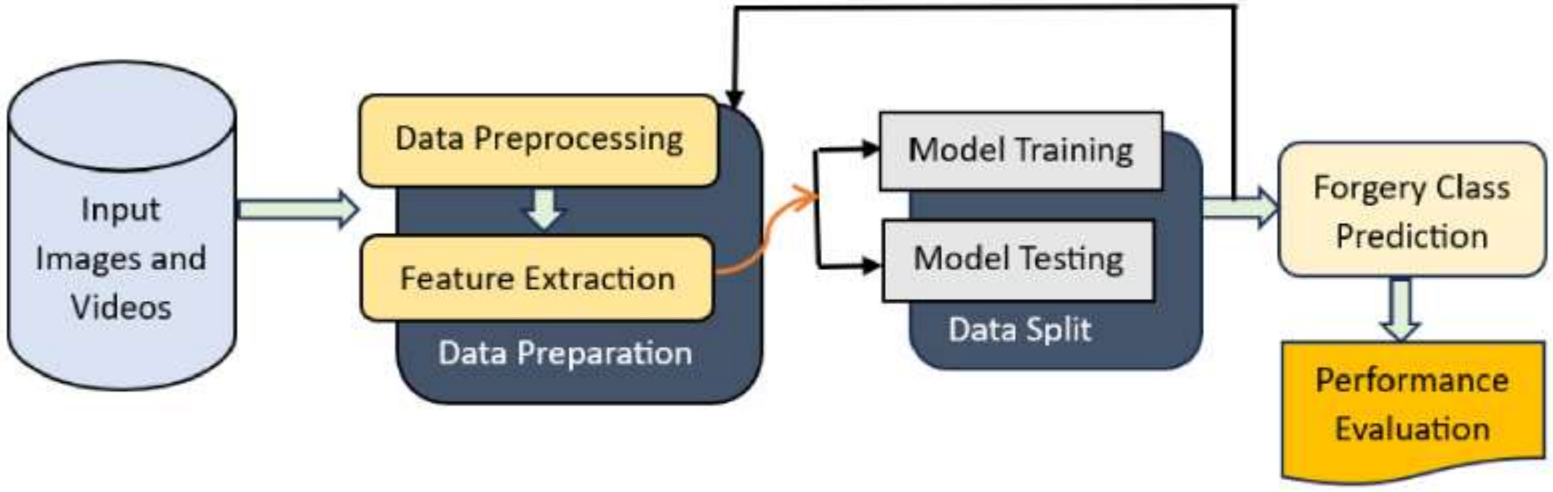
- This makes it difficult to know whether an image is real or manipulated.
- To solve this, we need smart systems that can automatically detect fake images.
- Image forgery can spread misinformation and cause security, media, and legal problems.
- Our project uses **Deep Learning** and **Generative AI** to identify forged images.
- The model not only detects if an image is fake but also shows the **exact area** where changes were made.
- This helps improve trust in digital content and supports fields like journalism, forensics, social media, and law enforcement.



An example of a copy-move image forgery; (a) is an original image, and (b) is a copy-move forged image [29]

Existing Approaches/ Related Work

- Earlier methods checked image noise, lighting, and sharpness to find edits.
- These traditional methods work only for basic or small forgeries.
- Machine learning models used manually selected features, but they were not very accurate.
- Deep learning models (like CNNs) improved detection by learning patterns automatically.
- Some methods try to highlight the edited area, but results are not always clear.
- GAN-based fakes are very realistic, and many old methods fail to detect them.
- Overall, existing methods can detect some forgeries but cannot reliably find where the image is edited or when the forgery is very advanced.

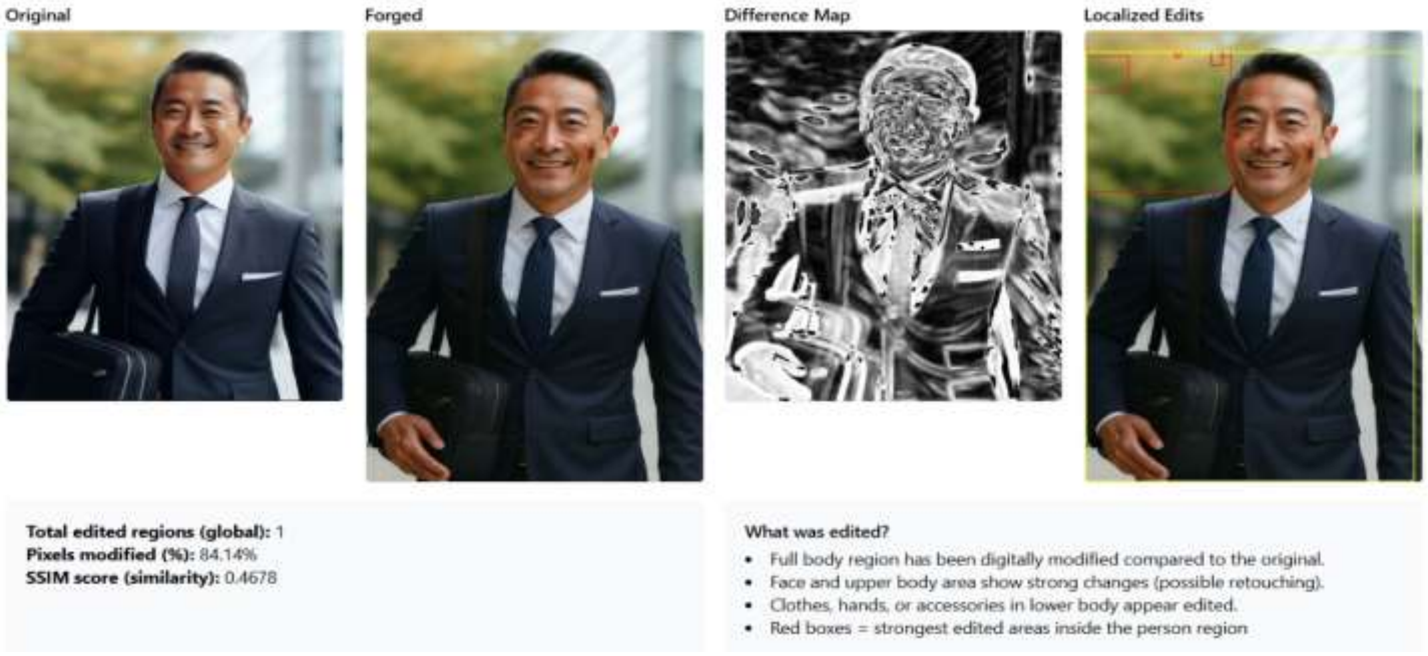


Problems in Existing Approaches

- Many old methods cannot detect complex or high-quality image forgeries.
- They fail when the edited part is very small or well blended.
- Most methods can say an image is fake but **cannot show the exact forged area**.
- Traditional models depend on manual features, which are not always accurate.
- Deep learning models still struggle with AI-generated images made by GANs.
- Many approaches work well only on specific datasets, not on real-world images.
- Some methods are slow and cannot be used for quick or large-scale detection.
- Forgery techniques improve fast, but many existing models do not adapt easily.

Research Objectives

•To develop a deep learning model that can accurately detect forged images.
•To identify and highlight the exact region of manipulation in an image.
•To improve detection accuracy for both traditional edits and AI-generated fake images.
•To create a robust system that works reliably on real-world images and different types of forgery.
•To use generative AI techniques to understand and expose advanced forgeries.



Proposed Methodology/Model

- Use a deep learning model(CNN) to analyze images and detect forgery patterns.
- Apply generative AI to learn hidden traces left by manipulated or AI-generated images.
- Use a segmentation model (like U-Net) to highlight the exact forged region.
- Train the model on real and fake images to improve accuracy and robustness.
- Evaluate the system to ensure it can both detect and localize image forgeries correctly.

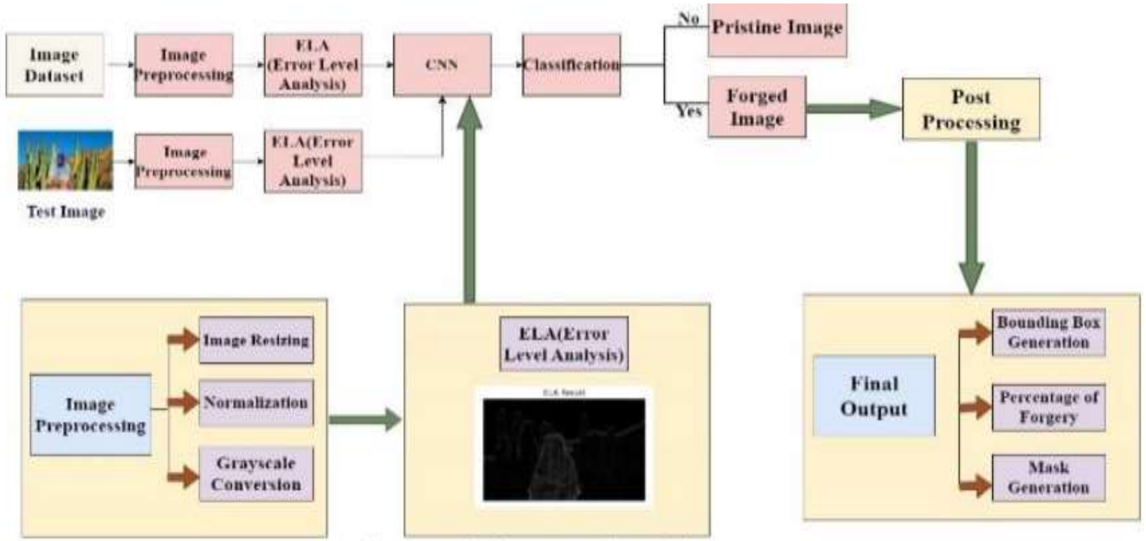
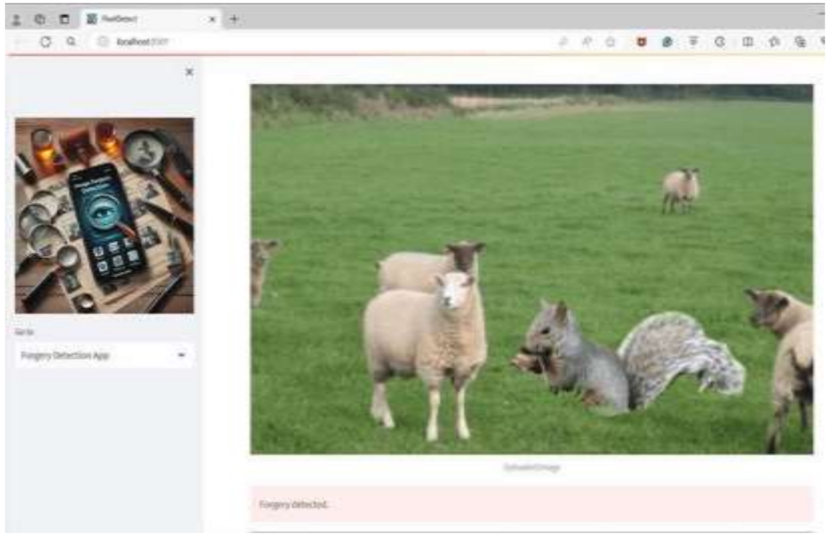
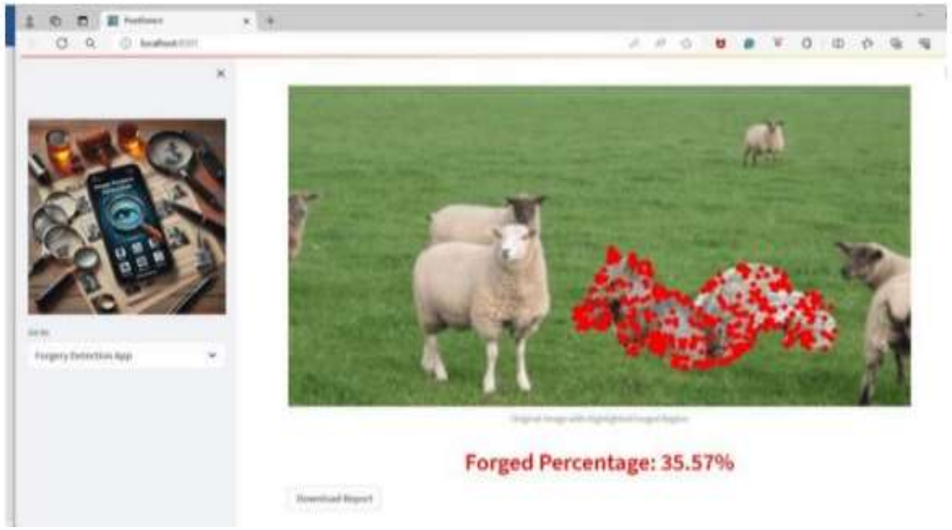


Figure 1 System Architecture

Results and Discussion

- The proposed model accurately identifies whether an image is real or forged.
- The system successfully highlights the exact manipulated region using the segmentation model.
- Deep learning and generative AI together improved detection performance compared to traditional methods.
- The model showed strong results on both normal edited images and AI-generated fake images.
- Experiments proved that the approach is more reliable and robust than existing methods.





Conclusion and Future Work

- The proposed deep forensic model effectively detects forged images and identifies manipulated regions.
- Combining deep learning with generative AI improves accuracy and handles advanced image forgeries.
- The system performs better than traditional methods and works well on different types of fake images.
- This model helps build trust in digital content by detecting both simple and AI-generated manipulations.

References

1. **IEEE Xplore Digital Library**
   Research papers on deep learning and image forgery detection.
   (Search: *“Image Forgery Detection using Deep Learning”*)
2. **ScienceDirect (Elsevier)**
   Journals covering digital forensics, CNN-based detection, and GAN forgeries.
   (Search: *“Deepfake detection”*, *“Image manipulation localization”*)
3. **SpringerLink**
   Articles on computer vision, image forensics, and segmentation models like U-Net.
   (Search: *“Image tampering detection using CNN”*)
4. **ResearchGate**
   Open-access research papers on generative AI and forgery detection techniques.
   (Search: *“GAN-based image forgery detection”*)
5. **arXiv.org (Cornell University)**
   Latest preprint papers on deepfake, GAN detection, CNN models, and forensic analysis.
   (Search: *“Image Forensics”*, *“Deepfake detection models”*)

THANK YOU